

Linear Models Lecture 17: Pooled and Panel Data

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Where We Are: From GMM to Panel Data

- In Lectures 15–16 we developed **GMM** as a unified framework:
 - Specify moment conditions $E[\mathbf{g}(\mathbf{z}_i, \theta_0)] = \mathbf{0}$
 - Estimate $\hat{\theta}$ by minimizing $\hat{\mathbf{g}}' \mathbf{W} \hat{\mathbf{g}}$
 - Test overidentifying restrictions with the J -statistic
- **Panel data** introduces new structure:
 - Repeated observations create natural moment conditions
 - Unobserved heterogeneity α_i must be controlled
 - Different assumptions $\Rightarrow$ different moment conditions $\Rightarrow$ different estimators
- **This lecture:** pooled OLS, fixed effects, between estimators as moment-based estimators.
Next lecture: random effects, CRE, and dynamic panel GMM (Arellano-Bond).

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Panel Estimators as Moment Conditions

Estimator	Moment Condition	Key Assumption
Pooled OLS	$E[\mathbf{X}'_{it} e_{it}] = \mathbf{0}$	α_i uncorrelated with $\mathbf{X}$
Fixed Effects	$E[\tilde{\mathbf{X}}'_{it} \tilde{e}_{it}] = \mathbf{0}$	Strict exogeneity (within-unit)
Random Effects	$E[\mathbf{X}'_{it} \nu_{it}] = \mathbf{0}$	α_i uncorrelated with $\mathbf{X}$; GLS weighting
Arellano-Bond	$E[Y_{is} \cdot \Delta e_{it}] = 0, s \leq t-2$	Sequential exogeneity; lagged levels as instruments

- Each row is a **GMM estimator** with different instruments and assumptions.
- More moment conditions $\Rightarrow$ overidentification $\Rightarrow$ testable with the J -test.

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From Cross-Sections to Repeated Observations

- A single cross-section identifies effects only by comparing units. Adding time lets us compare a unit to itself.
- Time can enter the model in several ways:
 - As an indicator (before/after a policy) or as a continuous trend.
 - Through interactions, allowing slopes to vary by period (the effect of a free fedora on employment may differ today from 1940).
- The simplest design exploits two groups at two times: **difference-in-differences**. We treat it as the on-ramp to panel data.

Differences in Differences

- We might collect data over time to rule out certain confounding factors.
- Consider a study of the effect (D) of a voter ID law on turnout (Y), observing Texas and Oklahoma before and after the law.

- Comparison one (cross-section, after the law):

State	Time	Outcome
Texas	After	$Y = M + T + D$
Oklahoma	After	$Y = O + T$

Difference = $(M - O) + D$: confounds the state-level gap with D .

- Comparison two (time-series, Texas only):

State	Time	Outcome
Texas	Before	$Y = M$
	After	$Y = M + T + D$

Difference = $T + D$: confounds the time trend with D .

- Neither comparison on its own recovers D .

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Potential Outcomes

- Define D as the binary treatment $d \in \{0, 1\}$ which tells us if we will be in the treatment category once we get to the second period.
- Measurements happen twice, $t \in \{0, 1\}$
- Y_t^d is the outcome in period t for treatment status d .
- The outcomes we can observe is Y_t (no d).

$$Y_t = dY_t^1 + (1 - d)Y_t^0$$

- Counterfactual outcomes:
 - $Y_t^1 | D_t = 0$ (Outcomes in a world of treatment among those who weren't)
 - $Y_t^0 | D_t = 1$ (Outcomes in a world of control among those who were treated)

$$\delta_i = Y_t^1 - Y_t^0$$

Parallel Trends

- Identification assumption/Parallel Trends

$$E[Y_{t=1}^0|D=1] - E[Y_{t=0}^0|D=1] = E[Y_{t=1}^0|D=0] - E[Y_{t=0}^0|D=0]$$

- In previous example, T is the same across states, state level differences ($M - O$) don't vary over time.
- The differences in expected potential non-treatment outcomes are unrelated to belonging to the treated or control group in the post treatment period.
- If the treated had not been subjected to treatment, both subpopulations would have experienced the same time trends.
- The composition cannot be affected by treatment

Obama Support: The Identification Problem

- Suppose we want to study the effect of racial co-identification (non-identification) on feeling thermometers of Obama in 2008.

$$ObamaRating_i = b_0 + d_{08}Black_i + \mathbf{b}'_x \mathbf{x}_i + e_i$$

- However, we are worried about the two kinds of omitted factors:
 - Time trends (T): Obama ran after a financial crisis and war that might make him popular generally.
 - Group Differences (M-O): Black and white people might differ in their preferences over any president that is a relatively liberal northern antiwar Democratic senator.
- Suppose Obama's appeal (repulsion) among Black (white) people depends on the economic conditions in 2008, his being a liberal Senator, and not only his race?

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Obama Support: DiD with a Comparison Candidate

- We can use a repeated cross section to difference out similar attitudes toward another Democratic candidate.

$$Kerry_i = b_0 + d_{04}Black_i + \mathbf{b}'_x \mathbf{x}_i + e_i$$

- If we assume that
 - All time varying factors other than the candidate's race affected Black and white people's response to Democratic nominees equally.
 - All differences in evaluations of D. nominees between Black and white people did not change between 04 and 08.
- Then δ is an estimate of the Average Treatment Effect of the candidate's race on attitudes.

$$\delta = d_{08} - d_{04} = (\bar{r}_{08,B} - \bar{r}_{08,W}) - (\bar{r}_{04,B} - \bar{r}_{04,W})$$

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Estimating Differences in Differences

- Stack 04 and 08 data. Define $yr08_i = 1$ for the 2008 survey takers and $B_i = \mathbb{I}(\text{Race} = \text{"Black"})$.
- **Unrestricted** (year-specific race effect):

$$DRating_i = b_{0,08}yr08_i + b_{0,04}(1 - yr08_i) + d_{08}yr08_iB_i + d_{04}(1 - yr08_i)B_i + \mathbf{b}'_x\mathbf{x}_i + e_i$$

- **Restricted** ($d_{08} = d_{04} \equiv d$):

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- The F test of $H_0: d_{08} = d_{04}$ comparing the two models is the DiD test: the racial co-identification effect is $d_{08} - d_{04}$.

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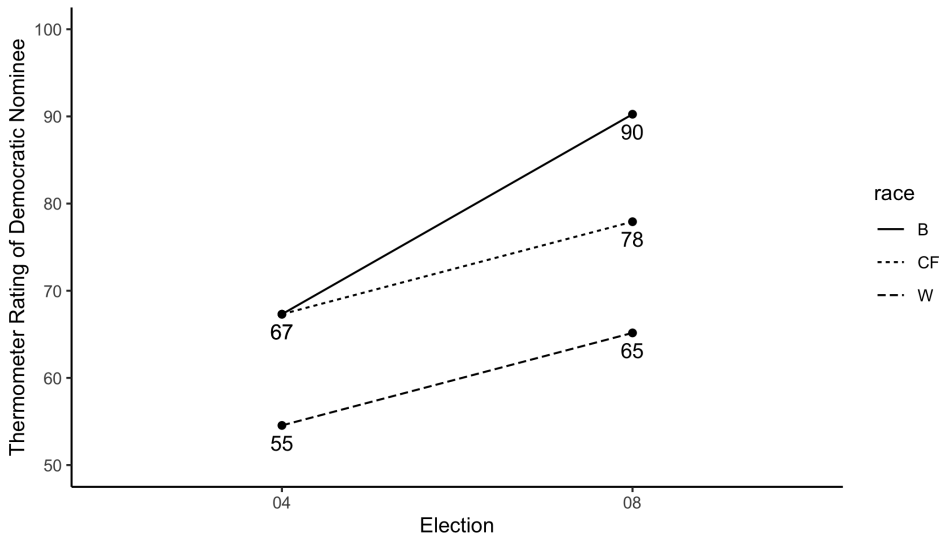
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	demft		
l(1 - y08)	54.5*** (1.4)	53.0*** (1.4)	
y08	65.2*** (1.3)	65.9*** (1.3)	10.6*** (1.0)
black		21.7*** (1.0)	12.8*** (1.9)
l(1 - y08):black	12.8*** (1.9)		
y08:black	25.1*** (1.2)		12.3*** (2.3)
age	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)
Constant			54.5*** (1.4)



Policy Evaluation

- Diff-in-diff is often used to study binary policies when we theorize that selection bias is driven by fixed group features.
- Key assumption for causal inference: parallel trends:
 - The amount of uncontrolled selection bias is not changing over time.
 - Time trends are the same across groups.
- Example of violation of parallel trends: Suppose that Texas and Oklahoma are trending blue, but Texas is closer to flipping, causing both voter ID laws and more investment by Democrats.

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Alternative: Synthetic Control

- When we are studying aggregates like states, we might not have parallel trends.
- However, in some cases we can construct a plausible counterfactual from combinations of untreated units.
- The synthetic control approach uses covariates to construct these counterfactual weighted combinations.
- Assumes that we have a long time series and the treated unit outcomes lie within the convex hull of untreated outcomes.

From Two Cells to Many: Panel Data

- DiD compared two groups at two times. Differencing cancelled time-invariant group features and group-invariant time trends.
- Panel data observes the *same* units repeatedly, so we can perform that cancellation for every unit, every period.
- Two costs of relaxing the two-cell setup:
 - Composition: groups must remain comparable across periods (no entry or exit driven by treatment).
 - Dependence: repeated observations on the same unit are correlated, breaking iid.
- The remaining sections build pooled, between, and within estimators on this richer structure.

- Each pairs a time-varying outcome Y_t , a time-varying regressor Z_t , and an unobserved unit-level feature A that we cannot measure directly.
- The same machinery (within transformation, GMM moment conditions) applies to all three, but the assumptions differ.
- As we proceed, ask of each example: which estimator handles it, and which would miss?

Panel Example 1: Voting behavior

- Consider a population of families. Choose one at random.
- For each family member t , we observe voting decisions Y_t and exposure to a persuasive messages Z_t .
- Each family has a household wealth W and a set of stories passed down from their grandparents A .
- We observe Y_t , Z_t and W , but not A .
- We want to estimate the effect of Z_t on Y_t , holding constant W and A .

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Panel Example 2: Airline cost structure

- Consider a population of airlines. Choose one at random.
- In each year, t , each airline pays Y_t dollars to fly Z_t passengers.
- Each airline uses a proprietary technology A which does not vary over time.
- We observe N randomly selected Airlines but do not measure A .
- We want to estimate the way that Y_t responds to Z_t , holding constant A .

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Panel Example 3: Candidate Electoral Performance

- Consider a population of candidates. Choose one at random.
- For each year t , the candidate receives vote share Y_t .
- We know each candidate has an underlying charisma A which does not vary over time.
- We observe N randomly selected candidates, but do not measure A .
- We want to estimate the effect of Y_{t-1} on Y_t , holding constant A .

What These Examples Motivate

- Voting in families: A (grandparents' stories) likely correlates with Z (messages); pooled OLS is biased and **fixed effects** restore comparability.
- Airline costs: A (proprietary technology) does not vary in time, so the **within** estimator removes it without measuring it.
- Candidate performance: Y_{t-1} appears on the right-hand side; the within transformation correlates with the error and **dynamic panel GMM** (next lecture) instruments with lagged levels.

Panel Data: Terminology

- The voters in a family, the airlines and candidates over time are all examples of *panel data*.
- Our observations are double indexed, usually divided into N units and T time periods/groups.
- If there are many T but few N , we call the data long and narrow, (time series cross section).
- If every unit is observed in every period/ group, the data are called balanced.

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Greene's Airline Panel

- Greene analyzes the cost structure in the airline industry from 1970-84.
- Research question: do airlines have positive economies of scale?
- `library(AER); data("USAirlines")`; analyzes determinants of total costs for 6 airlines.
- The cost function follows a Cobb-Douglas specification. Taking logs yields a linear regression:

$$\ln(\text{cost}) = \beta_0 + \beta_1 \ln(\text{output}) + \beta_2 \ln(\text{fuel price}) + \beta_3 \text{load} + u$$

- *cost*: total costs of labor materials etc in USD 1000.
- *output*: revenue passenger mile index
- input *prices*: fuel prices
- *load*: share of seats sold on flights flown.

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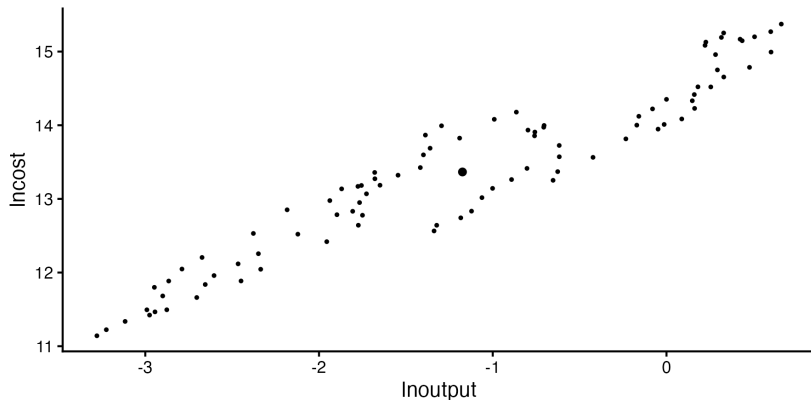
- *cost*: total costs of labor materials etc in USD 1000.
- *output*: revenue passenger mile index
- input *prices*: fuel prices
- *load*: share of seats sold on flights flown.

Pooled Regression of $\log(\text{costs})$

	Greene 2000	Greene 2003
(Intercept)	9.52*** (0.23)	9.42*** (0.23)
lnoutput	0.88*** (0.01)	0.94*** (0.03)
lnpfuel	0.45*** (0.02)	0.46*** (0.02)
load	-1.63*** (0.35)	-1.54*** (0.34)
lnoutput ²		0.02* (0.01)
R ²	0.99	0.99
Num. obs.	90	90

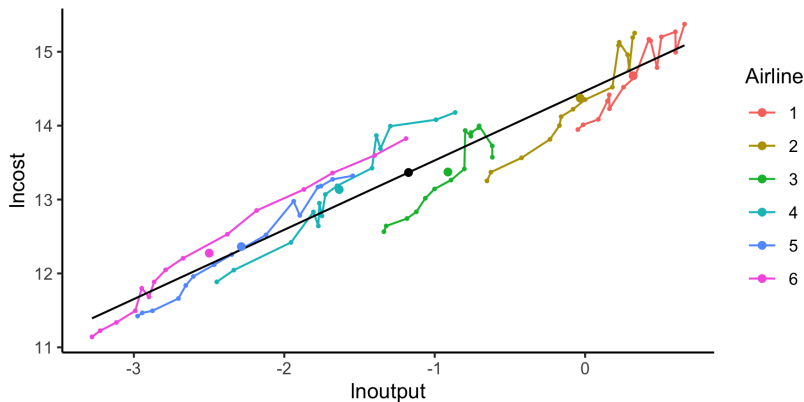
- Before reading off pooled coefficients, look at the data the way pooled OLS does: every airline-year is one point.
- Then strip off the airline-specific mean and ask what is left within an airline.
- The next plots compare those views for output, fuel price, and load.

Cost and Output: Raw Scatter



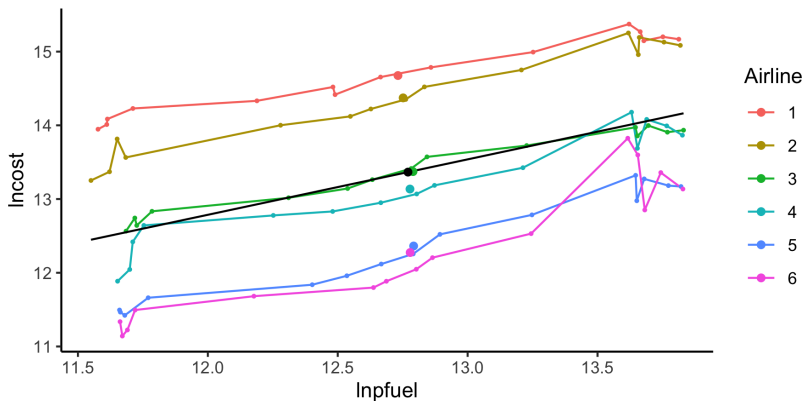
90 airline-years pooled. The black dot is the grand mean ($\overline{\overline{\ln \text{output}}}$, $\overline{\overline{\ln \text{cost}}}$), the only summary a single OLS fit will use.

Cost and Output: Pooled, Between, Within



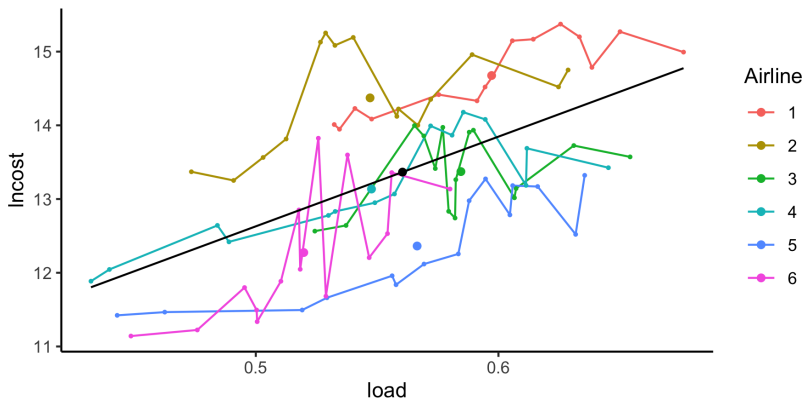
Coloured lines: each airline through time (within variation). Large coloured dots: airline means (between variation). Black line: pooled OLS fit.

Cost and Fuel Price



Within and between move the same way (prices rose over the 1970s for everyone), so the pooled slope is close to the within slope.

Cost and Capacity Utilization (load)



Bivariate pooled slope is mildly *positive*: airlines that run fuller planes also have higher total costs, because larger carriers do both.

What the Plots Show

- For output and fuel price, within and between push the same way, so pooled OLS looks fine.
- For load, between-airline variation reverses the sign relative to within. The bivariate pooled scatter is misleading.
- Partialling out the other regressors recovers the within-airline relationship. FE will do this for all variables at once.

Assumptions of Pooled Regression: Exogeneity

- Before turning to fixed effects, ask what pooled OLS needs to be unbiased once the same unit appears T times.
- $\mathbf{y} = \mathbf{X}\beta + \mathbf{e}$
- Strict Exogeneity: $E(e_{it}|\mathbf{X}) = \mathbf{0}$ for all time periods of $\mathbf{X}$.
- That is, $\mathbf{X}$ must not have any long-run consequences for $\mathbf{Y}$.
- Moreover, $\mathbf{X}$ must not itself be a function of past $\mathbf{Y}$.

Autocorrelation versus Autoregression

$$Y_t = \alpha Y_{t-1} + \beta X_t + \nu_t$$

$$\nu_t = \rho \nu_{t-1} + e_t$$

- $\rho \neq 0, \alpha = 0$ has a "common factor" problem (autocorrelation):
 - Outcomes at $t - 1$ are associated with outcomes at t for many unobserved reasons.
 - OLS slope estimates are unbiased but inefficient and produces wrong standard errors.
 - Addressed with GLS/ serial correlation robust standard errors.
- $\alpha \neq 0$ is associated with dynamics:
 - Outcomes at $t - 1$ affect outcomes now.
 - OLS is biased, inefficient and produces wrong standard errors.
 - Addressed with lagged variables and other structural time series approaches.

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Improvements on Inference

- In panel data we likely violate of iid errors, as observations from the same group/firm are likely associated

$$E[e_{ig}e_{jg}] = \rho\sigma_e^2 > 0$$

- Here ρ is the intra-group/firm correlation.

Standard error problems.

- If the number of groups is large, we can cluster standard errors using a formula similar to White standard errors.
- If the number of groups is small, we can just inflate our standard errors by $\sqrt{1 + (n_g - 1)\rho}$ (the Moulton factor; n_g is the (average) cluster size).
- There is a bootstrap solution that has good performance

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White vs Cluster Robust

	(white)	(cluster by airline)
(Intercept)	9.52*** (0.22)	9.52*** (0.38)
lnoutput	0.88*** (0.01)	0.88*** (0.02)
lnpfuel	0.45*** (0.02)	0.45*** (0.03)
load	-1.63*** (0.32)	-1.63* (0.44)
R ²	0.99	0.99
N Clusters		6

From Fixing Inference to Using the Panel Structure

- Clustered SEs adjust the variance of $\hat{\mathbf{b}}_P$ but leave the coefficient alone.
- The deeper problem is bias when α_i is correlated with $\mathbf{X}_{it}$: pooled OLS averages it in.
- Each unit is observed T times. Differencing each variable from its own unit mean cancels α_i without ever observing it.
- That move defines the within (fixed-effects) estimator. We derive it next, then show that pooled OLS is itself a weighted average of within and between.

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- Consider the typical panel

- $$\hat{\beta} = (X' M_D X)^{-1} X' M_D y$$

$$\hat{f} = (D' M_X D)^{-1} D' M_X v$$

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Within-group estimators

- Consider the typical panel model: $y_{it} = \gamma x_{it} + f_i + e_{it}$
- Assume that $E[e_i | f_i, x_{i1}, x_{i2}, \dots, x_{iT}] = 0 \quad \forall i$
- But suppose that $E(f_i | x_{it}) \neq 0$. We can dummy for "fixed effects" f_i , producing

$$y = X\beta + Df + e$$

- We know from Frisch Waugh Lovell that:

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- Where $\mathbf{M}_X = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\mathbf{M}_D = \mathbf{I} - \mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'$ are the idempotent residual makers.
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- $\hat{\beta}$ and $\hat{f}$ are the "within-group" estimators.

- Suppose $N=2$: $D = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}$ where i is a $(T \times 1)$ vector of 1s

$$\begin{aligned}
 M_D &= I - D(D'D)^{-1}D' \\
 &= \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} - \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \left[\begin{pmatrix} i' & 0 \\ 0 & i' \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \right]^{-1} \begin{pmatrix} i' & 0 \\ 0 & i' \end{pmatrix} \\
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Within Estimator

Two equivalent ways to estimate:

- 1 Explicitly controlling for fixed effects $\mathbf{f}$ and estimating $\hat{\beta} = (\mathbf{X}'\mathbf{M}_D\mathbf{X})^{-1}\mathbf{X}'\mathbf{M}_D\mathbf{y}$.
- 2 Differencing out the group means:

$$y_{it} - \bar{y}_i = \beta'(\mathbf{x}_{it} - \bar{\mathbf{x}}_i) + e_{it} - \bar{e}_i$$

$$\mathbf{b}_W = (\mathbf{S}_{xx}^W)^{-1}\mathbf{S}_{xy}^W$$

$$\mathbf{S}_{xx}^W = \sum_{i=1}^N \sum_{t=1}^T (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)'$$

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Stacked Matrix Notation

$\mathbf{y}_i: T \times 1, \mathbf{X}_i: T \times k, \mathbf{e}_i: T \times 1, \boldsymbol{\beta}: k \times 1$

$$\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_N \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_N \end{bmatrix} \boldsymbol{\beta} + \begin{bmatrix} \alpha_1 \mathbf{1} \\ \alpha_2 \mathbf{1} \\ \vdots \\ \alpha_N \mathbf{1} \end{bmatrix} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \vdots \\ \mathbf{e}_N \end{bmatrix}$$
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Stacked Matrix Notation

$\mathbf{y}_i: T \times 1, \mathbf{X}_i: T \times k, \mathbf{e}_i: T \times 1, \boldsymbol{\beta}: k \times 1$

$$\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_N \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_N \end{bmatrix} \boldsymbol{\beta} + \begin{bmatrix} \alpha_1 \mathbf{1} \\ \alpha_2 \mathbf{1} \\ \vdots \\ \alpha_N \mathbf{1} \end{bmatrix} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \vdots \\ \mathbf{e}_N \end{bmatrix}$$
$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\alpha} \otimes \mathbf{1}_T + \mathbf{e}$$

Software Implementations

- $\text{lm}(y \sim x + f_1 + f_2 + \dots + f_n)$
- $\text{plm}(y \sim x, \text{c}(\text{"unit"}, \text{"time"}))$
- $\text{fixest}::\text{feols}(y \sim x \mid \text{fe})$
- You can also demean by time averages, but you need to correct the standard errors.

Fixed Effects and unobserved factors

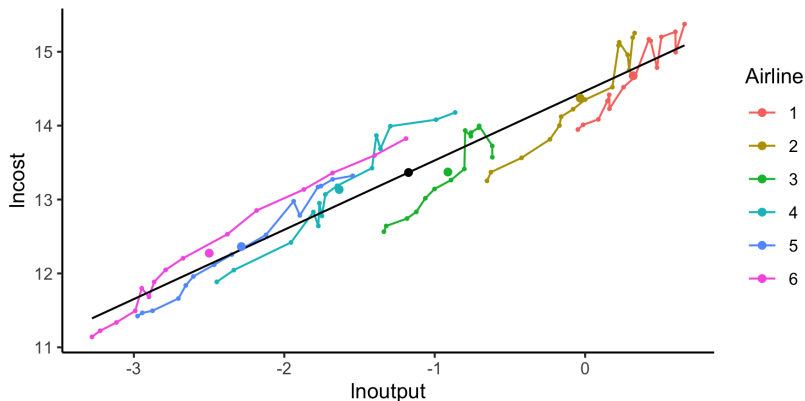
Just as with difference in difference, fixed effects address unobserved effects (α_i).

$$y_{it} = \gamma x_{it} + \alpha_i + e_{it}$$

The fixed effects transformation again mean deviates the variables, giving us:

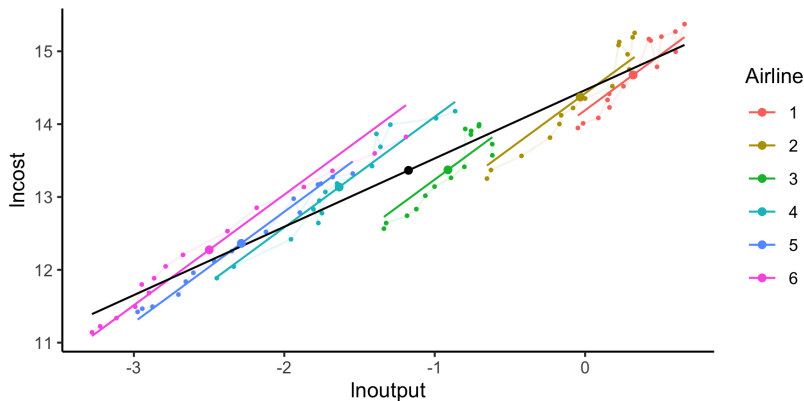
$$y_{it} - \bar{y}_i = \beta'(\mathbf{x}_{it} - \bar{\mathbf{x}}_i) + \alpha_i - \bar{\alpha}_i + e_{it} - \bar{e}_i$$

Cost and Output: Pooled Fit



Single black line: pooled OLS slope $\hat{\beta} = 0.88$, averaging across and within airlines.

Cost and Output: Within Fit



Per-airline fits with a common slope but separate intercepts. Within slope $\hat{\beta} = 0.92$ is steeper than pooled because the flatter between-airline relationship pulls the pooled line down.

Small note on estimation

$$\mathbf{b}_{fe} = \beta + (\mathbf{X}'\mathbf{M}_D\mathbf{X})^{-1}\mathbf{X}'\mathbf{M}_D\mathbf{e}$$

Under iid errors, we have: $\text{var}(\mathbf{b}_{fe}) = \sigma_e^2(\mathbf{X}'\mathbf{M}_D\mathbf{X})^{-1}$. But we need to correct our estimator for the N estimated means:

$$s_e^2 = \frac{\sum_{i=1}^N \sum_{t=1}^T e_{it}^2}{NT - N - K}$$

Fixed Effects as GMM Moment Conditions

- Define the within-transformed variables $\tilde{Y}_{it} = Y_{it} - \bar{Y}_i$ and $\tilde{\mathbf{X}}_{it} = \mathbf{X}_{it} - \bar{\mathbf{X}}_i$.
- The FE (within) estimator solves the moment conditions:

$$E[\tilde{\mathbf{X}}_{it}' \tilde{e}_{it}] = \mathbf{0}$$

- This is a **just-identified** GMM problem: k moment conditions for k parameters.
 - No overidentifying restrictions $\Rightarrow$ no J -test with FE alone.
- **Preview:**
 - Random effects adds between-group moments $\Rightarrow$ more efficient under stronger assumptions.
 - Arellano-Bond (next lecture) creates *many* moment conditions from lagged levels $\Rightarrow$ overidentification $\Rightarrow$ testable with J -statistic.

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Pooled Regression Anatomy

Rewrite pooled OLS in the notation we will reuse for between and within.

- Let $\mathbf{x}_{it}$ be the $k \times 1$ vector of regressors. Grand means: $\bar{\mathbf{x}} \equiv \frac{1}{NT} \sum_{i,t} \mathbf{x}_{it}$, $\bar{y} \equiv \frac{1}{NT} \sum_{i,t} y_{it}$.
- The pooled estimator: $\mathbf{b}_P = (\mathbf{S}_{xx}^P)^{-1} \mathbf{S}_{xy}^P$ with

$$\mathbf{S}_{xx}^P = \sum_{i,t} (\mathbf{x}_{it} - \bar{\mathbf{x}})(\mathbf{x}_{it} - \bar{\mathbf{x}})', \quad \mathbf{S}_{xy}^P = \sum_{i,t} (\mathbf{x}_{it} - \bar{\mathbf{x}})(y_{it} - \bar{y}).$$

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```
airlines<-airlines%>% mutate(  
  mlnoutput = mean(lnoutput),  
  mlnpfuel = mean(lnpfuel),  
  mload = mean(load))  
  
Xp<-airlines%>%transmute(  
  lnoutputdm = lnoutput-mlnoutput,  
  lnpfueldm = lnpfuel-mlnpfuel,  
  loaddm = load-mload)%>%  
  as.matrix()  
  
Sp_xx = t(Xp)%*%Xp
```


Between Regression

- Part of the variation in y_{it} and $\mathbf{x}_{it}$ is variation across units and part across time.
- The cross sectional part is just:

$$\frac{1}{T}(y_{i1} + y_{i2} + \dots + y_{iT}) = \beta' \left(\frac{1}{T}(\mathbf{x}_{i1} + \mathbf{x}_{i2} + \dots + \mathbf{x}_{iT}) \right) + \frac{1}{T}(e_{i1} + e_{i2} + \dots + e_{iT})$$

$$\bar{y}_i = \beta' \bar{\mathbf{x}}_i + \bar{e}_i$$

Define $\mathbf{b}_B = (\mathbf{S}_{xx}^B)^{-1} \mathbf{S}_{xy}^B$.

$$\mathbf{S}_{xx}^B = \sum_{i=1}^N \sum_{t=1}^T (\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})(\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})' = \sum_{i=1}^N T(\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})(\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})'$$

$$\mathbf{S}_{xy}^B = \sum_{i=1}^N \sum_{t=1}^T (\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})(\bar{y}_i - \bar{\bar{y}}) = \sum_{i=1}^N T(\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})(\bar{y}_i - \bar{\bar{y}})$$

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```
airlines<-airlines%>%
  group_by(airline)%>%
  mutate(
    gmlnoutput=mean(lnoutput),
    gmlnpfuel=mean(lnpfuel),
    gmload=mean(load))

Xb<-airlines%>%transmute(
  blnoutputdm = gmlnoutput-mlnoutput,
  blnpfueldm = gmlnpfuel-mlnpfuel,
  bloaddm = gmload-mload) %>%
  as.matrix()

Sb_xx = t(Xb)%*%Xb
```

```
airlines<-airlines%>%
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  mutate(
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    gmload=mean(load))

Xw<-airlines%>%transmute(
  plnoutputdm = lnoutput-gmlnoutput,
  plnpfueldm = lnpfuel-gmlnpfuel,
  ploaddm = load-gmload)%>%
  as.matrix()

Sw_xx = t(Xw)%*%Xw
```

Decomposition of Variance

$$y_{it} = \bar{y}_i + (y_{it} - \bar{y}_i)$$

$$s_{xx}^P = s_{xx}^B + s_{xx}^W$$

$$s_{xy}^P = s_{xy}^B + s_{xy}^W$$

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Pooled as a Weighted Average of Within and Between

$$\begin{aligned}
 \mathbf{b}_P &= (\mathbf{S}_{xx}^P)^{-1} \mathbf{S}_{xy}^P \\
 &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} (\mathbf{S}_{xy}^B + \mathbf{S}_{xy}^W) \\
 &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xy}^W + (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xy}^B \\
 &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W \mathbf{b}_W + (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^B \mathbf{b}_B \\
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 &= \lambda \mathbf{b}_W + (\mathbf{I} - \lambda) \mathbf{b}_B
 \end{aligned}$$

Three estimators on the same data, differing only in which variation they use. The next slide shows them side by side on the airline panel.

Pooled as a Weighted Average of Within and Between

$$\begin{aligned} \mathbf{b}_P &= (\mathbf{S}_{xx}^P)^{-1} \mathbf{S}_{xy}^P \\ &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} (\mathbf{S}_{xy}^B + \mathbf{S}_{xy}^W) \\ &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xy}^W + (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xy}^B \\ &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W \mathbf{b}_W + (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^B \mathbf{b}_B \\ &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W \mathbf{b}_W + (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W - \mathbf{S}_{xx}^W) \mathbf{b}_B \\ &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W \mathbf{b}_W + (\mathbf{I} - (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W) \mathbf{b}_B \\ &= \lambda \mathbf{b}_W + (\mathbf{I} - \lambda) \mathbf{b}_B \end{aligned}$$

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Pooled, Between, and Within: Airline Estimates

	Incost		
	Pooled	Between	Within
lnoutput	0.883*** (0.013)	0.782** (0.109)	0.919*** (0.030)
lnpfuel	0.454*** (0.020)	-5.524 (4.479)	0.417*** (0.015)
load	-1.628*** (0.345)	-1.751 (2.743)	-1.070*** (0.202)
Constant	9.517*** (0.229)	85.809 (56.483)	
Observations	90	6	90
R ²	0.988	0.994	0.993

What the estimates tell us

Reading the pooled column as Cobb-Douglas elasticities:

■ Output elasticity

$$\hat{\beta}_1 = 0.883 \text{ (pooled)} \implies \hat{r} = \frac{1}{\hat{\beta}_1} \approx 1.13.$$

The industry exhibits *mild increasing returns to scale* (costs rise less than proportionally with output).

■ Fuel price elasticity

$$\hat{\beta}_2 = 0.454 \implies \hat{\alpha}_F = \hat{r} \hat{\beta}_2 \approx 1.13 \times 0.454 \approx 0.51.$$

Roughly half of total variable cost is attributable to fuel.

- **Load factor effect** The coefficient on *load* (−1.63) means that a 1-percentage-point increase in capacity utilisation lowers total cost by about 1.6 %, consistent with spreading fixed operating expenses over more output.

From Cost Functions to Political Economy

- The airline panel illustrated the mechanics: removing unit heterogeneity, pooling within and between variation.
- The same machinery answers questions where the unit effect is the central concern, not a nuisance.
- Next: a country-year panel asking why top marginal tax rates rose so dramatically in the 20th century.

Political Science Application: Tax Progressivity and War

- **Scheve & Stasavage (2010, 2012):** Why did top tax rates rise dramatically in the 20th century?
- **Hypothesis:** Mass military mobilization during WWI/WWII created political pressure for progressive taxation as a “conscription of wealth.”
- **Data:** Country-year panel, 20 countries, 1850–1970.
 - Y_{it} : Top marginal income tax rate
 - X_{it} : War mobilization $\times$ high-mobility indicator
 - α_i : Country fixed effects (e.g. institutional differences, colonial legacies)
- **Why fixed effects?** Countries differ systematically in fiscal institutions, political systems, and economic structure. Pooled OLS would confound these with the mobilization effect.
- Strict exogeneity plausible: war mobilization driven by geopolitics, not anticipated tax policy.

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- Strict exogeneity plausible: war mobilization driven by geopolitics, not anticipated tax policy.

Scheve & Stasavage: R Implementation

```
library(plm); library(haven); library(fixest)
dta <- read_dta("progressTax.dta")
pdta <- subset(dta, year >= 1900 & year <= 1930)

# Fixed effects with plm
fe_mod <- plm(topratep ~ wwihighmobaft,
              data = pdta,
              index = c("ccode", "year"),
              model = "within")
summary(fe_mod)

# Equivalent with fixest (+ Newey-West SEs)
fe_mod2 <- feols(topratep ~ wwihighmobaft +
                 gdppcp + leftseatshp + munsuff + ratiop
                 | country,
                 data = pdta, panel.id = ~country + year,
                 vcov = NW(1) ~ country + year)
```

Summary

- Difference in difference cancels out group-invariant trends and time invariant group differences (whether of interest or not).
- Fixed effects give the same estimates as demeaning $\mathbf{X}$ and $\mathbf{y}$ by their over time means, the “within” estimator (so long as you correct the residual variance estimator).
- The within estimator corresponds to the GMM moment condition $E[\tilde{\mathbf{X}}_{it}' \tilde{e}_{it}] = \mathbf{0}$, and is just-identified.
- A “between” estimator only uses variation across units, averaged over time.
- Pooled Regression is a weighted average of the between and within estimates, with weights $(\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W$.
- **Next time:** Random effects adds between-group moments via GLS, and Arellano-Bond uses lagged levels as instruments in first-differenced dynamic panels. Both fit into the GMM framework from Lectures 15–16.